

Courts Without Borders

Given the upcoming Global Sports League (GSL), this study examines the essential factors in designing a fair and efficient scheduling model for an international sports league.

We begin by identifying the most suitable sport for GSL. **Basketball** is deemed the ideal choice due to its global reach, accessibility, and competitive landscape across different regions. Next, we select 20 teams on a global scale. To ensure an objective and balanced selection, we develop a ranking model based on the Analytic Hierarchy Process, incorporating three soft indicators: **team honor, popularity, and market value**. Additionally, **geographical representation** serves as a hard constraint to maintain diversity. After applying our model, we identify 20 teams from 16 countries across 6 continents.

With the participating teams established, the next crucial aspect is structuring the league itself. A successful league requires both a well-suited competition format and an efficient scheduling system.

To determine the most appropriate format for GSL, we compare existing, widely recognized sports formats (World Cup, Double Round Robin, Major League Soccer, and Swiss format) with our proposed **play-in-and-off format**, which integrates a preseason, a double-round robin group stage, a play-in phase, and a knockout stage. The evaluation is based on two quantifiable factors: **fairness and attractiveness**. As both factors exhibit stochastic variation, we employ Monte Carlo simulation to sample 10,000 seasons and compute their average values. These quantities are then aggregated using weighted criteria to assess the candidate formats. The results strongly support the play-in-and-off structure as the optimal format for GSL. Excluding pre-season, the championship-winning team will play between 17 and 28 games, with a total maximum of 127 games in the league.

In constructing a fair schedule for the group stage, we approach it as an optimization problem with two objectives: balancing the objectives of minimizing total travel distance and minimizing the differences between groups. We employed Genetic Algorithms to calculate and balance the condition of minimum total travel distance and equitable travel time between groups. For the 20 participating teams, our results indicate that a plausible estimation of minimum travel distance is 155,405 km, **representing a 57% optimization compared to the longest possible route**. The minimum standard deviation in travel distance across groups is 4034.44 km. A case study of the Los Angeles Lakers demonstrates the effectiveness of our model. Their total travel distance is approximately 28,190 km, which is relatively short and similar to that of another team in the same group, the Guangdong Southern Tigers, with a flight distance of 27,868 km. The knockout stage requires minimum optimization since every match is determined.

To ensure the schedule remains practical, we test it by planning specific timetables. Our scheduling principles include considerations for jet lag and equitable rest periods between matches. Finally, we compile our results into a comprehensive schedule table spanning nine months, detailing every game with specific dates and time slots.

Finally, we expand our model to 24 teams. We use the same team-selection model to include The simulation and ranking model for format validates play-in-and-off format as the best fit. The optimization model with six teams suggests that **total travel distance increases by 70039 km**. The standard deviation of the schedule increases by 2192.49 km, which is reasonable considering the addition of one team in each group.

Keywords: Genetic Algorithm, Sports Scheduling

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1 Introducing the GSL

1.1 Background

The International Multi-Continental Matchmaking Committee (IMMC) is launching the Global Sports League (GSL), a groundbreaking international team sports competition designed to balance fairness, geographic diversity, and practical logistics. The league will include 20 teams, with at least two representing each continent (excluding Antarctica), competing over an 8-to-9-month season. While the sport has not yet been finalized, the IMMC emphasizes the need for a team sport with global appeal and scalability. Teams must propose a scheduling model that ensures equitable matchups, minimizes travel distances, and promotes sustainability. The challenge also requires testing how the model adapts to league expansions and evolving constraints, ensuring the GSL remains fair, inclusive, and environmentally responsible as it grows.

1.2 Problem Analysis and Restatement

We are asked to build a model to create a schedule for a team sport's league. We restate the problem, and outline our approach as the following:

1. Select a team sport that can have global extension
2. Find out 20 teams that are globally diverse enough, at least 2 for each continent
3. Optimize factors of the schedule
4. Make up the final schedule

2 Sports, Teams and Formats

2.1 Choice of Sport

To begin with, we have to determine the sport to examine. In order to determine a *team sport*, we have to consider both geographic diversity and accessibility. The geographic diversity requires a sport to have representation across all continents, which eliminates certain regional sports like Cu Ju from consideration. On the other hand, the requirement of accessibility makes sports that require a certain level of development for a country impossible. For example, while esports have gained global popularity, their mechanisms' reliance on advanced infrastructure presents limitations. In some regions of high poverty rate, the lack of access to high-speed internet and advanced mechanisms makes it challenging to establish competitive esports teams and achieve GSL's target.

Under these conditions, we have chosen basketball as our sport of discussion. Its infrastructure can be built almost anywhere, and the game can be played in various settings, ranging from a simple hoop and ball to a full-scale venue. For example, even in poor countries like Angola, there are professional basketball teams. In short, due to the sport's widespread popularity and easy accessibility, basketball stands out as an excellent candidate for GSL matches.

2.2 Choice of Teams

We have chosen 3 soft indicators and 1 hard indicator for our team selection, being honors of teams, popularity of teams, geographic representation of teams and market value of teams.

Honors are chosen to represent the strength and the ability to attract fans of each team, for the stronger a team is, the more honor it will have, and the more fans it will attract.

$$H = 3N_{nba} + 2N_{EL} + N_{oth},$$

where H represents honor N_{nba} is the number of nba champions, N_{EL} is the number of Euro League champion and N_{oth} is the number of other champions.

Popularity is chosen to represent the ability of attracting people to watch the competition. A high popularity ensures enough income to achieve the goal of economic sustainability.

$$\Psi = \sum_s F_s, \quad s \in S$$

where Ψ represents popularity, S is the set of social media ¹ and F_s is the fan number on social media s . Note that for each team, we collect the sum of the top 3 social media with the largest annual active user count.

Geographic Representation acts as a hard constraint. It ensures that teams from every country can participate in the league.

Market Value of teams gives us the most direct and updated information on teams' strength. A team with a higher market value means that they are stronger in recent and past games. Part of the data is collected from Forbes[1], but for some teams that do not have market data like teams from Africa, we estimated their market value on a reasonable level.

After calculating the indicators for each team, we now use the Analytic Hierachy Method[2] to calculate weights for the indicators. The first step of the AHP method is to decide a Pairwise Comparison Matrix. We do so by assigning the relative importance using "common sense" and obtain the following:

	Honor	Popularity	Market Value	a_{ij}	Interpretation
Honor	1	$\frac{1}{2}$	$\frac{1}{6}$	1	i is equally important as j
Popularity	2	1	$\frac{1}{3}$	3	i is slightly more important than j
Market Value	6	3	1	5	i is strongly more important than j
				7	i is very strongly more important than j
				9	i is absolutely more important than j
				2, 4, 6, 8	Intermediate values for finer judgments

The indicators from top to bottom on the column and from left to right on the row are relatively honor, popularity, and market value. Since for every i, j, k there is $a_{ij}a_{jk} = a_{ik}$, the matrix has a CI of exactly 0, below 0.1, which is reasonable.

Next, we calculate the weights vector W through computing the eigenvector corresponding to the largest eigenvalue of the above matrix, and then normalize into percentages. Here is the result:

$$W = \{11.11\%, 22.22\%, 66.67\%\}$$

Thus, we are ready to score the teams. We first normalize each indicator by taking their arithmetic mean:

$$\tilde{V}_{ij} = \frac{V_{ij}}{\sum_k V_{kj}}, \quad i, k \in T, \quad j \in I,$$

¹including Facebook, Instagram, X, Youtube, Wechat, Weibo, Tiktok, Whatsapp

where T is the set of teams, I is the set of indicators, V_{ij} is team i 's original value of indicator j , and $\tilde{V}_{ij}$ is the normalized value respectively.

Finally, we calculate each team's score by

$$S_i = \sum_j \tilde{V}_{ij} \cdot W_j, \quad j \in I,$$

where S_i is the score of team i .

After giving the teams scores using AHP method, we picked teams with the following geographical constraints:

1. Each continent has at least 2 teams and at most 4 teams
2. Each country can contribute at most 3 teams
3. With the conditions 1 – 2 satisfied, pick the team with the highest score available

And after selection, the teams chosen for the leagues are in Table 1 . To represent their geographical locations, we plots the team in Graph 1

No.	Teams	Continent	Country	City	Score
1	Golden State Warrior	NA	USA	Los Angeles	6.43
2	Los Angeles Lakers	NA	USA	Los Angeles	5.96
3	Chigago Bulls	NA	USA	Chicago	4.05
4	Real Madrid	EU	Spain	Madrid	3.09
5	Toronto Raptors	NA	Canada	Toronto	2.42
6	FC Barcelona	EU	Spain	Barcelona	2.24
7	CSKA Moscow	EU	Russia	Moscow	1.90
8	Panathinaikos Athens	EU	Greek	Athens	1.00
9	Guangdong Southern Tigers	AS	China	Guangzhou	0.67
10	San Miguel Beermen	AS	Philipine	Manila	0.52
11	Liaoning Flying Leopards	AS	China	Shenyang	0.40
12	Flamengo	SA	Brazil	Rio de Janeiro	0.34
13	Al Ahly	AS	Egypt	Cairo	0.29
14	Perth Wildcats	OC	Australia	Perth	0.25
15	Boca Juniors	SA	Argentina	Buenos Aires	0.24
16	Sydney kings	SA	Australia	Sydney	0.17
17	Zamalek	AF	Egypt	Cairo	0.13
18	New Zealand Breakers	OC	New Zealand	Auckland	0.12
19	Petro de Luanda	AF	Republic of Angola	Luanda	0.11
20	AS Sale	AF	Morocco	Sale	0.07

Table 1: 20 Participating Teams

2.3 Candidates for Formats

Once the initial 20 teams have been successfully recruited, we must determine the league format to officially launch the GSL. We propose a 4-stage format named *Play-in-and-off*, and compare it against four existing formats adopted by well-known leagues.

2.3.1 A Word on Group Assignments

For all formats that involve a (randomized) group assignment, we follow the same principle. The 20 teams are first categorized into five pre-ranking tiers: Seed #1-4, #5-8, #9-12, #13-16, and #17-20. One team from each tier is then randomly drawn to form a group. As the first season, we use the scores for selecting teams to decide the seed rankings of the teams.

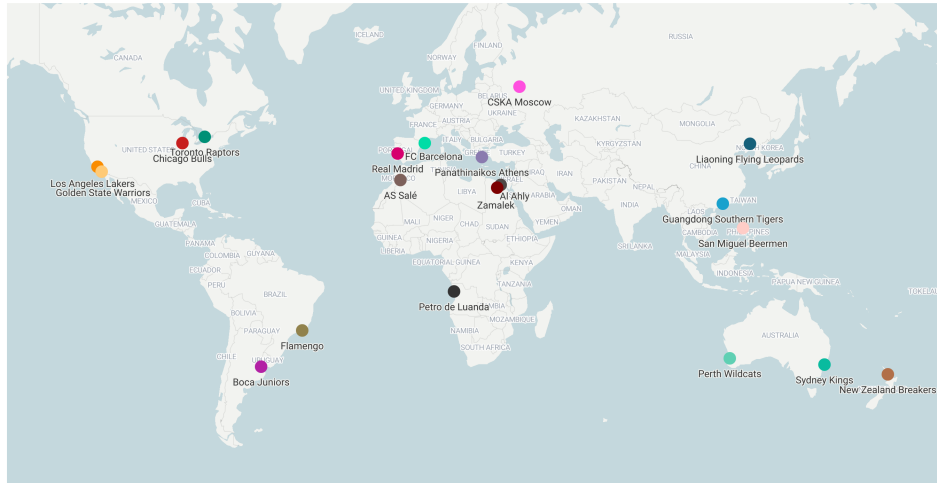


Figure 1: Geographic Distribution of the Teams

2.3.2 Existing Formats

Double Round Robin (DRR): In a DRR format, each team plays against every other team **twice**, one at home and one away. The final ranking will determine the season champion. This format is known for its efficacy [3], and we use it as a benchmark for evaluating the fairness of different formats[4].

Swiss format: The first stage is the group stage. 20 teams are divided into 4 individual groups, playing DRR in the group. The first four placed teams in each group will advance for the Swiss round. Then, the format is Swiss round. Using traditional Swiss format's rules [5], there are eight teams that have advanced ultimately. The eight advanced teams participate in the final elimination. The final champion will be determined by the finals[5].

World-cup Format: The first stage is the group stage. 20 teams are divided into 4 individual groups. The first two placed teams in each group will advance for the elimination stage. In the elimination stage, the final champion will be determined by the finals[6].

Major League Soccer(MLS)Format: The first stage is the group stage. 20 teams are divided into 2 individual groups. The first eight place teams in each group will advance to the elimination stage. In the elimination stage, all series are best of² one. All teams are randomly assigned their rivals. The final champion will be determined by the finals. There are fifteen games in the elimination in total[7].

2.3.3 Proposed Format: 4-stage Play-in-and-off

As previously outlined, we now propose a new format tailored for the GSL. The competition will consist of four sequential stages in the official season. The entire process of play-in-and-off schedule is visually represented in Graph 2.

Preseason: According to seed rankings, 20 teams are divided into 4 individual groups. In each group, there are 5 teams playing round-robin in their own group, 20 games in total. The preseason is only used for warm-up, not affecting any matches in the following stages. In addition, this stage will be played at a single arena, avoiding jet-lags. Preseason does not count for the official season.

The first stage is the group stage: In this stage, all five groups are playing DRR. Thus, there are $20 \times 4 = 80$ games, while each team will play 8 games at this stage. The higher the win rate, the higher the

²Best of N: The series has a maximum number of games of N. The team winning games more than half of the N is the winner of the series.

final rank the team gets. At the end of the group stage, five different final rankings will decide the formats teams are going to experience.

Ranking	Next Round
1 st place	Final Eight (Advanced to Stage 4)
2 nd place	Final Twelve (Advanced to Stage 3)
3 rd place	Play-in (Advanced to Stage 2)
4 th place	Play-in (Advanced to Stage 2)
5 th place	Eliminated

Table 2: Advancement after Stage 1

The second stage is the play-in stage: In this stage, only the 3rd and 4th placed teams compete for the remaining four spots in the Final Twelve, with all matches played as best-of-one (BO1) in third-party arenas.³ In the first round, the 3rd placed teams enter the upper bracket, while the 4th placed teams enter the lower bracket. Within each bracket, teams are randomly paired. The upper bracket winners secure a spot in the Final Twelve, while the lower bracket losers are eliminated. In the second round, the remaining teams face off: one upper bracket loser plays against one lower bracket winner. The winners advance, and the losers are eliminated. This stage consists of six matches in total: four in the first round (two per bracket) and two in the second round.

The third stage is the play-off stage: In this stage, eight teams advanced from both stage one and two compete for the remaining four seats of the Final Eight. All series in this stage are home-away series, best of three. Teams advanced from stage 1 hold home advantages. The series winners will advance for the Final Eight, and the others are eliminated. This stage has a maximum of twelve games.

The fourth stage is the final stage: The Series of the final eight, home-away series, best of three, are upcoming first. The teams that advanced from stage one have home advantages. The winners will become the candidates of the quarterfinals. Since quarterfinals, we will move to the third-party arena. The quarterfinals will be best of five, and the final will be best of seven. The champion will be decided by the outcomes of the final. This stage has a maximum of twenty-nine games.

Upgrading Seed Rankings: The previous season's final results are the seed ranking of each team for the next season. These are the seed rankings for the next season.

Previous Season	New Seed Ranking
Champion, Runner-up, Semifinals	Seed #1-4
Final Eight	Seed #5-8
Final Twelve	Seed #9-12
Play-in	Seed #13-16
Group Stage	Seed #17-20

Table 3: Corresponding New Seed Ranking

2.4 Evaluating Formats

In GSL, during eight to nine months, 20 teams will play matches and determine the final championship, and each team will be ranked by an initial seed ranking, which indicates the skill level of the teams. The better the team plays, the higher seed ranking they get. We use a linear rating system to quantify the overall ability

³ Third-party Arena: The arena that is placed in the continents except the continents where both teams come from.

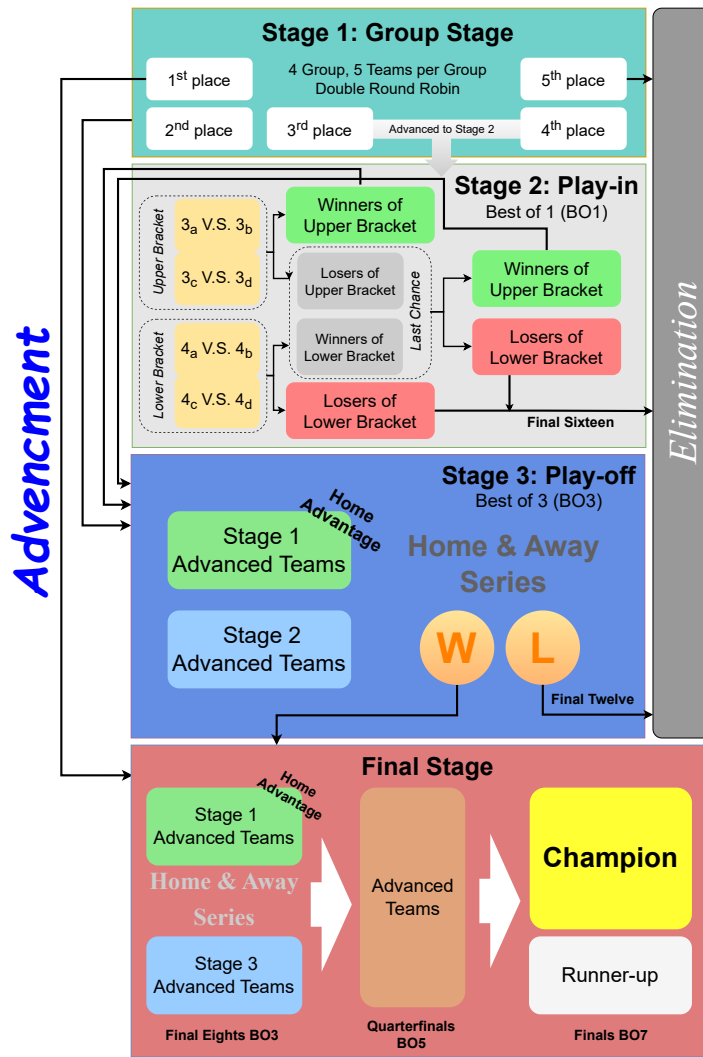


Figure 2: Flow Chart of Play-in-and-off Format

of each team. The variable, r_i , represents the ability of i seed ranked team, which should be measured by a positive value. The defined equation is:

$$r_i = 21 - i \geq 0, \quad i = 1, 2, 3, \dots, 20$$

Meanwhile, a higher seed ranked team is rated as a better performance, which is consistent with the definition. Then, we should introduce an equation to simulate the winning probability of each team. We use the *Bradley-Terry* model to simulate the possibility that team i wins team j [8].

$$P(\text{team}_i \text{ wins}) = \frac{r_i}{r_i + r_j}$$

During simulations of seasons, the result of each individual game is only determined by winning possibilities, not affected by the format itself. It is worth noting that these are only the generated data purely for evaluating the efficacy of various formats. Thus, they can provide fair comparisons and determine the most effective Format.

We use two factors to measure a format: **fairness and attractiveness**. The following section will delve into each one.

Fairness We evaluate format fairness through the *Top-4 Difference*, since we especially concern the final standing of the top teams. We take the absolute value of the expected rankings and final outcome of each team. *Top-4 Difference* is the average value of the differences of top four teams.

When using a pure double round-robin format, this calculation is straightforward since teams are ranked. However, in formats with knockout rounds, we classify teams into **brackets** based on their final placement in the tournament, such as "Champion," "Finalist," "Semifinalist." We define each bracket's expected placement as the **median** of all possible ranks within that stage.

For example, in the **World Cup** format, the eight remaining teams form the final eight (quarterfinals) bracket, while the final four teams advance to the semifinal bracket. The following table defines the brackets and their respective median values in the World Cup format.

Bracket Stage	Median
Champion (1st)	1.0
Runner-up (2nd)	2.0
Semifinalist (3-4th)	3.5
Final Eight (5-8th)	6.5
...	...

Table 4: Bracket classification for World Cup Format.

Thus, if a team was seeded 4th (expected bracket = Top 4, median = 3.5) but finishes in the Top 8 (actual median = 6.5), its bracket difference is calculated as $|6.5 - 3.5| = 3$.

Attractiveness In order to account for how many audiences a match can attract, we define A_{ij} , to be the ability to gain fans of a match of team i and j . Two indicators are used to calculate the attractiveness in a match of team i and j , being Q_{ij} , the quality of a match and C_{ij} , the competitiveness of a match. We define the quality of the game as the following according to[3]:

$$Q_{ij} = r_i + r_j$$

[3] remind here that r_i and r_j represents seed ranked to team i and j . This is to say that the higher the strength of teams are, the higher the quality of a match is.

Slightly adjusting equations in [3], we define competitiveness as the following:

$$C_{ij} = 1 - \frac{|r_i - r_j|}{D_{max}} \quad (1)$$

where D_{max} is the max difference between team ranks, 19 in the case of 19 teams. This shows the greater the difference in strengths of the teams, the lesser the competitiveness. This makes sense because a greater difference in team strengths reduces the likelihood of closely contested matches.

Now that we get the value of the two indicators, it's time to calculate the attractiveness of a match. We calculate attractiveness using the following formula according to[3]:

$$A_{ij} = Q_{ij}C_{ij}$$

This formula shows attractiveness is proportional to both quality and competitiveness. This means that for attractiveness to be high, both quality and competitiveness must also be high. If either one is too low, the overall attractiveness will decrease. To reflect this relationship, we combine the two indicators using multiplication.

2.4.1 Simulation Process

The following is a summary of how we conduct each simulation using *Python*.

1. **Team Setup.** We have n teams, labeled by a seed i from 1 to n . Each team's rating is given by $n + 1 - i$, and this rating remains fixed throughout the simulation.
2. **Group Assignment.** We randomly assign teams into groups according to the rules described in Section 2.3.1 (e.g., two groups of 10, or four groups of 5, depending on the format). This step is skipped in a pure double-round-robin across all 20 teams (Premier League Format), as group assignments are not needed.
3. **Match Definition.** Based on the chosen format, we list out every match that needs to be played. For instance, some formats involve only group play, while others have group stages followed by knockout rounds. BO1 and BO3 games are also defined.
4. **Match Simulation.** For each match, we use the Bradley–Terry model [8] to calculate the probability of one team defeating the other, then generate a random outcome (no ties). We also compute an *attractiveness* score based on both teams' ratings. If a team wins, we update its progress (e.g., advancing in a bracket or increasing its win count).
5. **Outcome of a Season.** After all matches are finished, we determine each team's final placement. In a round-robin, teams are sorted by total wins (ties broken using the same probability of winning). In a knockout, we track who advances until a champion is decided. We then calculate the *Top 4 Difference* using the bracket-based ranking system described in the Fairness (Section 2.4), where expected placements are determined by median rankings.
6. **Iteration.** Steps 1 to 5 are repeated for 10,000 seasons to account for randomness and provide stable statistical results.

2.4.2 Simulation Results and Analysis

After running 10,000 simulations for each format, we tracked two key factors: fairness and attractiveness. Fairness is measured by the Top 4 difference score (Table 5) and the bracket distribution of the first seed. Attractiveness is measured by the overall attractiveness score for the season (Table 6).

Below, we break down the results for each factor.

A. Fairness We first analyze fairness, which we quantify using the Top 4 Difference and the #1 Seed Team distribution.

1. Top 4 Difference Table 5 shows the average difference of top 4 seed teams. View this as a "standard error" between the predicted outcome (seed number) and simulated rank. A smaller difference score means better fairness and stability in results. Among all formats, double-round-robin had the smallest average difference, making it the most stable. Aside from that, MLS and Play-in-and-off also showed low differences between expected and actual results. A result that might seem counter-intuitive is the high value top 4 difference for the Swiss format, which is usually recognized for its fairness and stability. This happened because of a modification in our simulation—normally, in knockout rounds, higher-seeded teams play against lower-seeded ones. But in our model, we **randomized** these matchups. This meant some top teams faced stronger opponents earlier than expected, leading to a higher difference score.

2. First Seed Distribution Now that we've examined the Top 4 Difference, another important aspect of fairness is how the first seed performs. Specifically, we look at its placement across different formats to see if the strongest team consistently earns top rankings. Thus, we tracked its performance and created a histogram (see Figure 3).

For ranks with no set final outcome, we assume equal probability within the respective bracket.

Format	Seed 1	Seed 2	Seed 3	Seed 4	Average
DRR	2.44	2.44	2.44	2.46	2.44
MLS	3.34	3.12	2.82	3.20	3.12
Play-in-and-off	3.37	3.49	3.35	3.67	3.47
Swiss	5.24	4.89	4.47	4.67	4.82
World Cup	5.46	5.12	4.76	4.90	5.06

Table 5: Top 4 Difference of All Formats(Ascending)

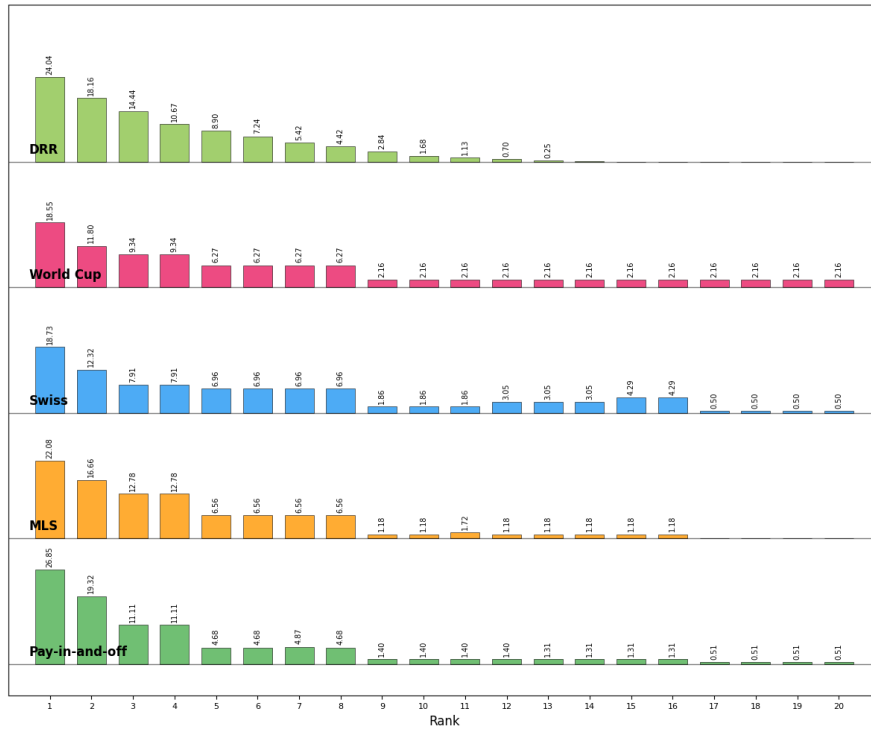


Figure 3: Final Rank Distribution of the #1 Seed Team Across All Formats

In Figure 3, every histogram is skewed to the right. This makes sense—since the first seed is the strongest (having the highest rating), it has a higher chance of ranking at the top. In the play-in-and-off format, the first seed has the highest probability of winning the championship. This is likely because the number of games in later knockout rounds increases from BO3 to BO7, making it more likely for the higher-ranked team to win.

In the double-round-robin format, the first seed is most likely to finish in the top 4. The distribution steadily decreases, and there is zero chance of the first seed finishing below 13th place. This pattern reinforces the stability of double-round-robin. Nevertheless, double round-robin is not considered due to its prolonged season and a lack of attractiveness for most games, as explained in section 2.4.

Because double-round-robin is recognized for its fairness, we use it as a benchmark. **We compare the first seed's histogram in other formats to that of double-round-robin to see how similar (or different) they are.** A distribution similar to that of double-round-robin is deemed to have better fairness. To measure this, we perform Jensen-Shannon (JS) Divergence for our 5 distributions, which improves upon Kullback-Leibler (KL) Divergence. The KL Divergence for distributions P and Q is defined as follows:

$$D_{KL}(P||Q) = \sum_i P(i) \log_2 \frac{P(i)}{Q(i)}$$

The smaller the value, the more similar the distributions. However, KL Divergence has two key issues: it approaches infinity if $\exists i, Q(i) = 0$, and it's not symmetric, meaning $D_{KL}(P||Q) \neq D_{KL}(Q||P)$. Therefore,

we use an upgraded version-JS Divergence. JS Divergence is defined as the following:

$$D_{JS}(P||Q) = \frac{1}{2}D_{KL}(P, M) + \frac{1}{2}D_{KL}(Q, M)$$

where $M = \frac{(P+Q)}{2}$. In this way, for every i , $M(i) > 0$, so $\frac{1}{M(i)} \neq \frac{1}{0}$, and the divergence is symmetric for P and Q . Moreover, this divergence has a bounded value between 0 and 1 for every single term, compared to the unbounded value of KL Divergence. Like the KL Divergence, the closer $D_{JS}(P||Q)$ is to 0, the more similar P and Q are. For example, when $P = Q$, $M = P = Q$, the JS Divergence is 0. And for two Bernoulli distributions with success probability $P = 100\%$ and $Q = 0\%$, $D_{JS}(P||Q) = 1$.

To evaluate fairness across formats, we calculate $D_{JS}(t, drr)$ for every $t \in T$, where T represents the rank distribution of the first seed in all selected formats (including double-round-robin). The results are shown in the middle row of Table 6.

Since DRR is being compared to itself, its JS Divergence value is 0—there's no difference. Other formats show varying degrees of deviation from DRR's distribution. Among them, play-in-and-off is the closest with a divergence score of 0.42, while Swiss and World Cup formats have the highest divergence at 0.53.

B. Attractiveness Another crucial aspect we evaluate is attractiveness. As shown in Table 6, the play-in-and-off format ranks highest in attractiveness, followed by Swiss and World Cup.

Looking at the data, we notice a general trend: formats with higher attractiveness tend to have a higher Top 4 Difference, suggesting a trade-off between fairness and attractiveness. This pattern, discussed in [3], makes sense. Greater fairness usually means stronger teams facing weaker ones early on, reducing competitiveness and making games less attractive. However, play-in-and-off stands out as an exception. Despite its high attractiveness score, its Top 4 Difference isn't as extreme. This is likely due to the BO7 and BO5 series, where highly competitive teams play in later rounds. These extended series increase the average competitiveness of games, helping to maintain excitement while still allowing stronger teams to perform well.

2.4.3 Ranking Format

Now that we've looked at fairness and attractiveness separately, it's time to choose the best overall format. However, no single format ranks consistently highest in all three indicators, making it difficult to pick a clear winner. So instead of choosing the "best" in one category, we need to find the format that offers the best balance between fairness and attractiveness. To solve this, we developed a ranking model that evaluates each format based on both factors.

Normalization Before we move on to the ranking calculation, note that the three factors are measured on different scales. Since their raw scores aren't directly comparable, we need to normalize each factor to be a value between 0 and 1.

The attractiveness score is a positive indicator—higher values mean a better format. Because the data has low variance, we use max normalization, dividing each value by the maximum. In math expression:

$$\bar{A}' = \frac{\bar{A}}{\max(\bar{A})}$$

This ensures all values are scaled between 0 and 1, with the most attractive format (play-in-and-off) getting a score of 1.

The other two indicators, Top 4 Difference and JS Divergence, are negative indicators—smaller values mean a better format. For Top 4 Difference, we use reciprocal normalization to ensure that smaller values receive

higher normalized scores:

$$\Delta T'_4 = \frac{1}{\Delta T_4}$$

This ensures that formats with smaller bracket differences get a higher normalized score.

However, JS Divergence requires a different approach. Since double-round-robin (DRR) has a JS Divergence of 0, reciprocal normalization would cause division by zero. To address this, we slightly modify the max normalization method used for attractiveness. We have:

$$JS' = 1 - \frac{JS}{\max(JS)}$$

We invert them so that a lower original value results in a higher normalized score. DRR, with a JS value of 0, gets the highest normalized score of 1.

These normalization methods allow us to compare all three factors on the same scale, making them suitable for the final ranking calculation.

Weight Generalizations With all three factors now normalized to a common scale, we can move on to the next step: assigning weights to reflect their relative importance in the ranking model.

We set the fairness-to-attractiveness ratio at 4:6. Since GSL is a new league, the main priority is to attract fans and drive engagement to cover high costs, increase profits, and grow its popularity [9]. Because of this, attractiveness carries more weight in our ranking model. This means the weight for *attractiveness score* is 0.6, and the combined weights for *top 4 difference* and *first seed distribution* is 0.4.

We further break down the fairness weight. As defined in Section 2.4, the #1 Seed Team's placement plays a major role in fairness. Since Top 4 Difference already considers the first seed's ranking, both First Seed Distribution and Top 4 Difference inherently measure fairness for the strongest team. Given this overlap, we assign equal weights of 0.2 to both factors.

Therefore, we have the expression for final format score as:

$$\text{Schedule Score} = 0.6\bar{A}' + 0.2JS' + 0.2\Delta T'_4 \quad (2)$$

Result After calculation, our format score is recorded in Table 6.

Format	Top 4 Difference Avg	JS Divergence	Avg Attractiveness	Format Score
Play-in-and-off	3.47	0.42	20.17	0.697
DRR	2.44	0.00	13.26	0.676
Swiss	4.82	0.53	15.33	0.498
MLS	3.12	0.47	13.27	0.482
World Cup	5.06	0.53	14.84	0.481

Table 6: Format Rank (Descending)

The top-ranked format is play-in-and-off, mainly because of its high attractiveness score. Since attractiveness carries the most weight in our ranking model, it's no surprise that this format comes out on top.

Right behind it is double-round-robin (DRR), which scores the highest in both fairness measures but ranks lower overall due to its low attractiveness score. This highlights the trade-off in DRR—it prioritizes fairness at the cost of excitement and engagement.

Since play-in-and-off has the best overall score compared to other formats, it is considered the most suitable format for GSL. Moving forward, we will use play-in-and-off for the rest of our model.

3 Develop the Initial Schedule

After finalizing the format for GSL, the next step is to confirm the schedule in group stage. That is, we then determine the match schedule for the group stage.

3.1 Objectives

For the "play-in-and-off" format, we structure the group stage as a double round-robin tournament, considering two key factors: total travel distance across all teams and fairness in travel distribution. Our goal is to strike a balance between minimizing overall travel and ensuring teams have similar travel demands.

Minimizing total travel distance is important for two reasons. First, shorter trips allow teams more rest between games, helping them maintain peak performance. Second, reducing travel cuts down on carbon dioxide (CO₂) emissions, aligning with the tournament's sustainability goals.

However, focusing solely on total travel can lead to unfair schedules, where some teams travel much more than others. Excessive travel reduces recovery time, putting those teams at a disadvantage. To address this, we also aim to minimize the standard deviation of travel distances, making schedules more equitable.

Since these two objectives—reducing total travel and ensuring fairness—cannot be optimized simultaneously, we use a weighted approach to balance them. Additionally, a standard double round-robin tournament requires an even number of teams, so we make a necessary assumption to accommodate this in our calculations.

Assumption 1

For each group, we adjust the format by giving one team a **bye** (sitting out) each round. Since each team plays every other team twice, the total number of matches is $2 \times \binom{5}{2} = 20$. With only 4 teams (maximum even number below 5) playing per round, each round consists of 2 matches. Thus, each group requires 10 rounds. Thus, **each team sits out exactly twice throughout the tournament.**

Before we proceed to the details of our objective function and its constraints, we define the following parameters and binary variables notations used. This procedure applies to one group, and we will repeat it for all four groups.

- n : Number of teams. This is 5 as discussed above. Individual teams are indexed by $i, j, t \in \{1, \dots, n\}$.
- r : Number of rounds. This is 10 as discussed above. Since we have 5 teams, each team needs to play $4 \times 2 = 8$ games and thus 8 rounds. Individual rounds are indexed by $k \in \{1, \dots, 2n - 2\}$.
- d_{ij} : Distance from team i 's home venue to team j 's home venue. We record this by collecting a 20×20 distance matrix, with distance determined by the cities of the teams.
- The binary variable x_{ijk} , y_{tijk} and z_{ijk} are defined as:

$$x_{ijk} = \begin{cases} 1 & \text{if team } i \text{ plays } \mathbf{away} \text{ against team } j \text{ in round } k. \\ 0 & \text{otherwise.} \end{cases}$$

$$y_{tijk} = \begin{cases} 1 & \text{if team } t \text{ travels from the team } i \text{ venue to the team } j \text{ venue between rounds } k \text{ and } k + 1. \\ 0 & \text{otherwise.} \end{cases}$$

$$z_{ijk} = x_{ijk}, \quad i, j = 1, \dots, n : i \neq j, \quad k = 1, \dots, 2n - 2,$$

$$b_{ik} = \begin{cases} 1, & \text{if team } i \text{ has a bye in round } k, \\ 0, & \text{otherwise.} \end{cases}$$

Varibale y means team i should be at the home city of team j if the former plays away against the latter. Essentially, once the opponent and venue are determined by x_{ijk} , this variable helps answer the question: Where is team i scheduled to play in round k ? (This variable might seem redundant at first sight, but it simplifies later constraints.)

To incorporate the minimization of the standard deviation of travel distances alongside the objective of minimizing total travel distance, we use a weighted sum approach. Thus we have:

$$\min \alpha \cdot \text{Total Travel Distance} + \beta \cdot \text{Standard Deviation of Travel Distances}$$

where α is the weight for total travel distance and β is the weight for standard deviation. Total travel distance is given by:

$$\sum_{t=1}^n \sum_{i=1}^n \sum_{j=1}^n \sum_{k=0}^{2n-2} d_{ij} \cdot y_{tijk} \quad (3)$$

This function can be broken down into three stages: 1. Each team travels from its home city to the location of its first game. 2. Teams move between game locations for the remaining rounds 3. If a team finishes the tournament away from its home city, it must travel back. Thus, the travel distance can be rewritten as:

$$\sum_{i=1}^n \sum_{j=1}^n d_{ij} \cdot x_{ij1} + \sum_{t=1}^n \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^{2n-3} d_{ij} \cdot y_{tijk} + \sum_{i=1}^n \sum_{j=1}^n d_{ji} \cdot x_{ij,2n-2} \quad (4)$$

Another objective is to minimize the **standard deviation** of travel distances across teams. Let $D_t = \sum_{i,j,k} d_{ij} \cdot y_{tijk}$ denote the total travel distance for team t . Let $\bar{D} = \frac{1}{n} \sum_{t=1}^n D_t$ be the mean travel distance. Standard deviation is expressed as:

$$\text{Standard Deviation} = \sqrt{\frac{1}{n} \sum_{t=1}^n (D_t - \bar{D})^2} \quad (5)$$

Combining objective 4 and 5, we have our objective function.

$$\min \alpha \left(\sum_{i,j} d_{ij} x_{ij1} + \sum_{t,i,j,k} d_{ij} y_{tijk} + \sum_{i,j} d_{ji} x_{ij,2n-2} \right) + \beta \cdot \sqrt{\frac{1}{n} \sum_{t=1}^n \left(D_t - \frac{1}{n} \sum_{t=1}^n D_t \right)^2} \quad (6)$$

For this objective function, we deem both objectives as equally important: we can not overlook one at the expense of another. Therefore we assign equal weights of 0.5 to α and β . Next, there are many restraints we need to specify to mirror the reality of a double round-robin tournament. In fact, developing constraints in this case is similar to specifying the rules of traveling for teams. To facilitate the comprehension of these complex expressions, we will start by specifying small parts that build on to later expressions.

- $\sum_{j=1}^{2n-2} x_{ijk}$: This expression sum over all rounds $2n - 2$, which represents the total occurrences of a **specific game** (e.g., team j hosting team i) across the group stage.
- $\sum_{j=1}^n x_{jik}$: This expression sums over all opponents j , which represents **total home games for team i in a specific round k , which is 0 or 1**. The expression can be further altered, such as changing round number to $k + 1$ ie. $\sum_{j=1}^n x_{ji,k-1}$, which represents the total home games in the next round.

Thus, we identify the constraints first by referring back to the definition of double round-robin tournament (our adjusted version), which entails the following:

$$\sum_{k=1}^{2n-2} x_{ijk} = 1, \quad \forall i, j = 1, \dots, n, \quad i \neq j$$

This ensures that each team will play away against each opponent exactly once. Taking reversely, the same restraint also guarantees that each team will play exactly one home game against each opponent as well. To ensure that no teams can play against themselves, we have:

$$x_{iik} = 0, \quad i = 1, \dots, n, \quad k = 1, \dots, 2n - 2,$$

Next, after slight adjustment on the constraints of traditional double-round-robin, we have:

$$\sum_{j=1}^n (x_{ijk} + x_{jik}) + b_{ik} = 1, \quad \forall i = 1, \dots, n, \quad k = 1, \dots, 2n - 2$$

This means each team in every round either plays exactly one game (away or home, $\sum_{j=1}^n (x_{ijk} + x_{jik} = 1)$) or has a bye ($b_{ik} = 1$). However, not every team can have a bye at the same time. According to Assumption 1, in every round only one team has the privilege for a bye, which directly leads to:

$$\sum_{i=1}^n b_{ik} = 1, \quad \forall k = 1, \dots, 2n$$

Additionally, across the entire group stage, each team has exactly two byes, also according to Assumption 1. This is represented as:

$$\sum_{k=1}^{2n-2} b_{ik} = 2, \quad \forall i = 1, \dots, n$$

After developing the constraint of x , b and y variables, we define the logical relationship between y and z (or x) variables. We will do this in chronological order (i.e. following the sequence of a tournament). *Also, we will break them down into cases, though the constraints hold regardless of whether cases are met*

In the first game (i.e., first round), if team t travels from its home city to the city of team i in the first round, we do the following setup to represent the initial trip.

$$y_{tti1} \geq x_{it1} \quad \forall t, i \neq t.$$

From round 2 to the last round, there are two cases. We will use team i when talking about home game in the current round and t when talking about away game in the current round.

Case 1. The team i plays a home game in the current round k . That is, we set the location of i in its own venue.

$$z_{iik} = \sum_{j=1}^n x_{jik}, \quad i = 1, \dots, n, \quad k = 1, \dots, 2n - 2$$

Note that $\sum_{j=1}^n x_{jik}$ is 1 when i plays at its own venue in round k . Then,

1) If the next game is a home game, the venue of i is unchanged and thus no action needed.

2) If the next game is an away game against team j , then team i travels from its home city to that of team j .

$$y_{iijk} \geq \sum_{j=1}^n x_{jik} + x_{ij,k+1} - 1 \quad \forall t, i \neq t; \quad k = 2, \dots, 2n - 2. \quad (7)$$

As noted before, $\sum_{j=1}^n x_{jik}$ represents the total home games of team i in round k , which should be 1 if the current game is a home game. In this condition both $\sum_{j=1}^n x_{ji,k-1}$ and $x_{ij,k-1}$ (team i plays an away game against team j) are 1, so the left part should be 1.

Case 2. The team t plays an away game in the city of i in the next round. 1) If the next game is a home game for team t , it travels from the city of i back to the home city. Similar to (7),

$$y_{titk} \geq z_{tik} + z_{tt,k+1} - 1 \quad \forall t, i \neq t; \quad k = 2, \dots, 2n - 2.$$

2) If the next game is an away game for team t at the city of j , it travels from the city of team i to that of j .

$$y_{tijk} \geq z_{tik} + z_{tjk+1} - 1, \quad \forall t, i, j = 1, \dots, n, \quad k = 1, \dots, 2n - 1.$$

However, to proceed with the above constraint, we made an important assumption:

Assumption 2

If a team plays two **consecutive away games**, meaning they compete at one away venue in a given round and then at a different away venue in the next round, they travel directly from the first venue to the second without stopping at any other location in between.

After the last round, team t needs to return to its home city if it plays away in the last round.

$$y_{tit,2n-2} \geq x_{ti,2n-2} \quad \forall t, i \neq t.$$

We add another constraint during the group stage, which is based on our assumption.

Assumption 3

We avoid scheduling back-to-back matches between the same two teams to keep the competition fair and engaging. Consecutive matches could give one team an unfair advantage, especially if they have momentum from a recent win [10]. Additionally, this also helps distribute rest periods and match difficulty more evenly across teams.

$$x_{ijk} + x_{jik} + x_{ij,k+1} + x_{ji,k+1} \leq 1, \quad i, j = 1, \dots, n, \quad k = 1, \dots, 2n - 2,$$

Finally, variables x, y, z , and b are all binary.

$$x_{ijk}, z_{ijk}, y_{tijk}, b_{ik} \in \{0, 1\} \quad t, i, j = 1, \dots, n, \quad k = 1, \dots, 2n - 2.$$

3.2 Optimizing Solutions with Genetic Algorithms

Given these competing objectives and combinatorial constraints, we require an optimization approach that could handle all the constraints and optimize our schedule. We built a genetic algorithms model that could meet all the needs. To provide a clearer understanding of our model, the following pseudocode presents a detailed description of each step.

Our model improves the schedules using two key operations. The first one is crossover, which combines parts of two parent schedules. Selected schedules undergo crossover, which means two schedules are combined to produce new ones:

$$x_{\text{new}} = \text{Crossover}(x_p, x_q)$$

Algorithm 1 Genetic Algorithm for Tournament Scheduling Optimization

Input: Teams T , Distance matrix D , Rounds n_r , Parameters MAX_{bye} , POP_{size} , GEN_{num} , P_{cx} , P_{mut}

Output: Optimal schedule S^* minimizing total distance and variance

function INITIALIZEINDIVIDUAL

$M \leftarrow \text{permutations}(T, 2)$ ▷ All possible matches

Randomly shuffle M **return** M

end function

function EVALUATEFITNESS(*individual*)

$S \leftarrow \text{DecodeSchedule}(\textit{individual})$ ▷ Transform matches to rounds

if not ValidConstraints(S) **then return** (∞, ∞)

end if

$dist \leftarrow \text{CalculateJourney}(S, D)$ ▷ Calculate team distances

$total \leftarrow \sum_{t \in T} dist[t]$ ▷ Total distance

$fairness \leftarrow \sigma(dist.values())$ ▷ Standard deviation

return $(total, fairness)$ ▷ Multi-objective fitness

end function

▷ Main Genetic Algorithm

▷ Minimize both objectives

Create fitness function with weights $(-1.0, -1.0)$

Create individual type with list representation

$P \leftarrow \text{Initialize population of size } POP_{size} \text{ using } \text{InitializeIndividual}()$

Configure genetic operators:

Two-point crossover with probability P_{cx}

Shuffle mutation with probability P_{mut}

NSGA-II selection for multi-objective optimization

for $g = 1$ to GEN_{num} **do**

$offspring \leftarrow \text{Select parents from } P$

Apply crossover to $offspring$ with probability P_{cx}

Apply mutation to $offspring$ with probability P_{mut}

Evaluate fitness of $offspring$ using EvaluateFitness()

$P \leftarrow \text{Select next generation from } P \cup ofspring \text{ using NSGA-II}$

end for

$pareto_front \leftarrow \text{Extract non-dominated solutions from } P$

$S^* \leftarrow \text{Decode best schedule(s) from } pareto_front$

return S^*

where x_p and x_q are two parent schedules. We apply two-point crossover, meaning two random points k_1 and k_2 are selected, and segments between them are swapped:

$$x_{\text{new}} = (x_p[1 : k_1], x_q[k_1 : k_2], x_p[k_2 : L])$$

where L is the length of the schedule.

On the other hand, an individual may undergoes mutation to introduce small random changes:

$$x_{\text{mut}} = \text{Mutate}(x_{\text{new}})$$

We use swap mutation, where two positions i and j are randomly chosen and swapped:

$$x_{\text{mut}}[i], x_{\text{mut}}[j] = x_{\text{mut}}[j], x_{\text{mut}}[i]$$

Alternatively, other small modifications, like adjusting home/away status or adding rest days, can be applied.

After using these arithmetics, we could get a set of optimal solutions. To refine the selection, we filter solutions where the standard deviation of travel distances remains within an acceptable limit:

$$\sigma_{\text{travel}}(x) \leq \sigma_{\text{max}},$$

Limiting the standard deviation to the maximum flight distance in a single day ensures that all teams, regardless of their travel distance, can fully recover by the next day. If the travel distance disparity exceeds one day's worth of flight, teams with longer journeys effectively lose a full day of rest compared to others,

creating an unfair competitive disadvantage. We find out that the average speed of an airplane is 800 km/h and the maximum allowable standard deviation $\sigma_{\max}$ is set to the average maximum flight distance in a single day:

$$\sigma_{\max} = 800 \times 24 = 19200 \text{ km.}$$

Finally, the optimal schedule is determined by minimizing the total journey distance while satisfying the standard deviation constraint. The calculation outputs the selected schedule.

3.3 Result

The genetic algorithm produced a Pareto front consisting of several feasible scheduling solutions. Among these, we selected the optimal schedule based on our defined criteria: the standard deviation of travel distances must not exceed the maximum one-day flight distance (19,200 km), and from those schedules, the one with the shortest total travel distance was selected. **The selected optimal solution achieved a total travel distance of 155,405.00 km, with a standard deviation of 4034.44 km.**

To see how well our genetic algorithm works, we compared it to the worst possible travel schedule—the one with the longest total travel distance, which is 361,444 km. Our optimized schedule cut that down to 155,405 km, reducing the total distance by 57.0

On top of that, the variation in travel distances (standard deviation) was 4,034.44 km, which is well within the acceptable range. This big improvement from the worst-case scenario shows that our genetic algorithm is doing a great job at optimizing both travel efficiency and fairness.

In order to show the optimized routes, we here show an example of the route of the team Los Angeles Lakers in stage 1:

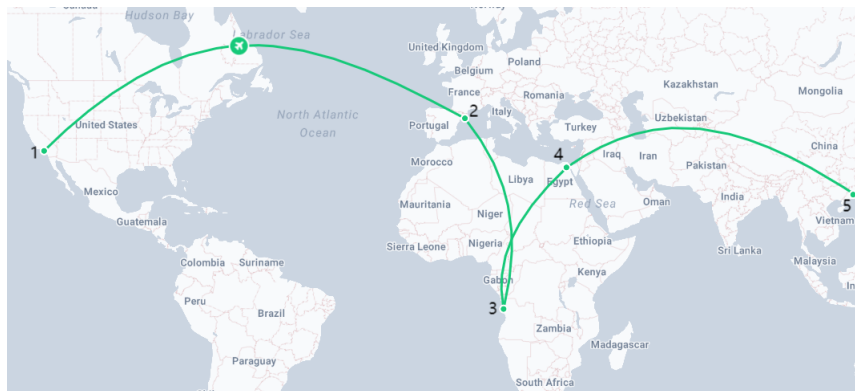


Figure 4: Los Angeles Lakers's Stage 1 Route Graph

On their way from LA to Luanda, the Lakers first stop in Barcelona. This route is actually shorter than flying to Luanda first and then to Barcelona because it follows a path closer to the direct curve from LA to Luanda. Taking routes like this helps keep the total travel distance as short as possible.

To check if the travel schedule is fair, we can compare the total flight distances of the Lakers and the Guangdong Southern Tigers. The Lakers travel 28,190 km in total, while the Southern Tigers travel 27,868 km—a difference of just 322 km. Since this only adds about half an hour of flight time, we consider it an acceptable difference.

3.4 Season Schedule

After knowing the order of matches, we scheduled the time of matches according to the following principles:

1. Every team must rest for at least 2 days after a match.

2. A team that has just got down from the plane needs to rest a day more
3. The standard deviation of every team's the average resting time is as small as possible with conditions 1 and 2 satisfied.

As for the result, it is shown on the 6.2. In our time schedule, there is a rest on every Monday. In this way both the teams and fans can have a day to take their rest. And for every team, their time schedule satisfies our conditions. And as an example, the time schedule for team Flamengo in stage 1 is shown below. As we can see here, its minimum days of rest is 9 days long. This gives them enough time to rest for the next game.

25.10.8	25.10.25	25.11.11	25.11.21
vs Boca Juniors	vs Real Madrid	vs CSKA Moscow	vs Real Madrid
25.11.29	25.12.10	25.12.19	26.1.13
vs Zamalek	vs CSKA Moscow	vs Zamalek	vs Boca Juniors

Table 7: Flamengo Schedule

4 Adding Four Teams

4.1 Choice of Teams

Now that we need four more teams in our league, the priority is to choose them out. We continue to use the ranking model we used before to choose our new teams. However, with only four teams available this time, we need to have a stricter constraint to ensure the global representativeness.

In order to do so, we have chosen four teams under the following new constraints: First, each Continent can have at least 1 new team joined. If previous condition is satisfied, choose the teams with the highest score from the rest of the teams. Therefore, there are 4 new teams:

1	Boston Celtics	NA	USA	Boston	3.99
2	Maccabi Tel Aviv	EU	Israel	Jerusalem	1.14
3	Al Riyadi Beirut	AS	Lebanon	Beirut	0.28
4	São Paulo FC	SA	Brazil	São Paulo	0.16

Table 8: 24 Participating Teams

After adding the new teams, the league is more diverse on a global scale. Two new countries, Israel from Europe and Lebanon from Asia, now join the league. Moreover, the new teams all come from different cities than the previous teams.

4.2 Format Analysis

Expanding the league from 20 to 24 teams not only broadens its geographical representation but also affects the suitability of the formats discussed in 2.3.2. The addition of new teams will inevitably lead to slight modifications in these formats. To evaluate how well the formats adapt to this expansion, we decided to apply the comprehensive ranking model introduced in 2.4.3 with 24 teams.

Note that changes in the number of teams will also affect the ranking system and the probability expressions for match outcomes. First, the seed number i is now indexed from 1 to 24, leading to an updated rating formula:

$$r_i = 25 - i \geq 0, \quad i = 1, 2, 3, \dots, 24$$

Additionally, in the competitiveness expression of a competition 2.4.2, the maximum possible difference between team ranks increases to $25 - 1 = 24$.

The format itself also undergoes slight modifications. However, this adjustment is relatively straightforward, as most existing tournament structures naturally accommodate 24 teams rather than 20. In particular, for the World Cup, the traditional 24-team format remains unchanged, eliminating the need for any adjustment. In the ranking, we use the same normalization method and weight distribution as 2.4.3. Results are demonstrated in Table 8. From Table 8, we can see that as the number of teams increases to 24, the

Format	Top 4 Difference Avg	JS Divergence	Avg Attractiveness	Format Score
Play-in-and-off	3.83	0.15	26.36	0.798
DRR	2.74	0.00	15.94	0.636
MLS	3.51	0.13	15.94	0.580
World Cup	5.48	0.27	18.84	0.496
Swiss	6.05	0.29	18.44	0.453

Table 9: Format Ranking with 24 Teams

top 4 difference also grows. This makes sense—when there are more teams, the probability of any single team winning the championship naturally decreases. At the same time, having more teams in each group makes the DRR results (whether in group stage or pure DRR) more robust, meaning strong teams are less likely to be unexpectedly eliminated. This is reflected in how the first-place distribution across all formats aligns more closely with DRR. Ultimately, despite these changes, the play-in-and-off format still achieves the highest overall format score, confirming that it remains the most suitable choice even with an increased number of teams.

4.3 Optimization Solutions

When expanding the schedule to a six-team double round-robin format, we modified our genetic algorithm to adapt to the increased complexity. Specifically, we removed the bye-round mechanism since an even number of teams allowed all teams to compete in each round without needing byes. This adjustment simplified the schedule structure, as every team participates in every round, eliminating the constraint management associated with bye rounds. The genetic algorithm's parameters, such as population size and mutation and crossover probabilities, remained the same to maintain consistency.

Here is one example of our result. In Group 1, the optimized total travel distance was 225,444.00 km, which increased by 70039.00 km compared to the total distance before adding the additional team. The standard deviation increased to 6,226.93 km, which is an increase of 2192.49 km. We could observe a small increase in total travel distances and fairness disparities. This increase is primarily attributed to the additional team, resulting in more matches and, consequently, higher cumulative travel, so we consider this result reasonable.

5 Generalizing to Other GSL Sports

Our tournament model has a lot of potential for use in different GSL sports, thanks to its well-designed structure. One of its biggest strengths is the grouping system, which ensures that each team competes against a mix of stronger and weaker opponents. This balance keeps the competition fair and engaging. In cases where a new tournament doesn't have pre-existing rankings, our model's approach—using pre-season results as a ranking reference—can help set up fair groupings right from the start.

Another major advantage of our model is its ability to guide the planning of international tournaments. By applying our structure, organizers can make better decisions about which countries and continents should host matches. Since our model also compares different existing tournament formats, similar sports can adopt

it with minimal adjustments.

That said, different sports have their own unique characteristics, so adjustments are sometimes necessary. One major consideration is the rules of each sport. Take football (soccer), for example. Unlike some sports, football matches can end in a draw, which means group-stage games need a points-based system instead of a win/loss record. Another key difference is match length. A standard football game lasts 90 minutes, which is significantly longer than a basketball game. Because of this, formats like best-of-three (BO3) aren't ideal for football, and scheduling back-to-back matches could cause fatigue and lower performance. Studies on athlete recovery times suggest that intense physical activity like football requires longer breaks between matches compared to sports with shorter game durations [11].

Apart from game rules, tournament planning must also consider logistical and technical factors. Esports, for instance, introduce entirely different challenges. Network issues, game crashes, or bugs can interfere with matches, sometimes requiring a rematch to keep things fair. An example is the 2022 League of Legends Mid-Season Invitational, where differences in internet latency led to match replays [12]. Another challenge in esports is frequent game updates. Since patches can change gameplay mechanics, tournaments need clear policies on whether updates will apply mid-event or if a single version of the game will be used throughout [13].

Traditional sports also face logistical challenges, such as weather conditions affecting schedules. Football matches, for instance, may need to be rescheduled due to extreme weather, requiring flexible planning to ensure player safety and match fairness. Unexpected delays can impact team performance and tournament pacing. In the end, while our tournament model is strong and adaptable, successful implementation requires sport-specific adjustments.

6 Conclusion

6.1 Strengths

Competitive Tournament Our game has chosen teams with the highest strength when possible. This ensures that the games have fierce competition, making the games more interesting.

Broad Application We have quantified indicators for every part, so our model could be applied on other projects, with a different sport type or different size.

6.2 Weakness

Complexity The format of our league is too complex. This could cause difficulties for fans catching up with the games, and could cause preparation of the games to become harder.

Incomplete Considerations of Routes When we are optimizing the routes of games, we didn't quantify factors other than distance, such as crossover effects. This could lead to certain problems like stronger teams are more likely to perform better than weaker teams.

Incomplete Considerations of Team Strengths When we compare different formats, we concluded our results based on our selection of teams only. However, when the strength distribution of the teams changes, the performance of different formats might change as well, but we failed to check up for that. This might limit the applicable range of our model.

Designing Schedule for the Global Sports League

Team #25069749

Honored Members of the IMMC,

This is team 25069749 writing to you about our model of deciding the schedule of the GSL. Thank you for having confidence on our team, and we are determined to give out a schedule that satisfies all the IMMC goals.

After careful analysis, we have produced a shcedule that satisfies all 4 goals of golabal representation, fairness, economic sustainability and environmental sustainability.

To achieve global representation, we used certain constraints in the selection of teams to make sure that every continent and country can have nearly equal amounts of teams joining in.

Fairness is achieved through the careful selection of fotmat of the league. We considered the fairness over several formats, and the format is made sure to let the stronger teams have higher possibility of goaling, making sure the game is has a rule fair to strength. Moreover, in the section of making time and route schedule for teams, we optimized the schedule so that each team as nearly equal amount of resting time and travelling distance.

Meanwhile, we also considered the attractiveness of games to achieve goal of economic sustainability. In the selection of formats, we accounted attractiveness as a major factor. This makes sure that our schedule attracts enough fans to gain economic support, making economic sustainable development possible.

Last but not least, we achieved environmental sustainability through optimizing the travelling distance of the teams. We made sure that the total travelling distance is minimized, so that the pollution caused through travelling is minimized, ensuring environmental friendly league.

In summary, our model harmonizes the IMMC's objectives, delivering a competitive, equitable, and sustainable framework for the GSL. We look forward to discussing how this approach can be adapted to future expansions or evolving priorities.

Sincerely,

Team #25069749

Pre-Season Stage

Aug.4	5 LAL/GST	6 ZLK/MCF	7 SDK/LFL	8 NZB/SMB	9 GST/AAL	10 MCF/FMG
11	12 LFL/PTN	13 SMB/ASS	14 AAL/PLD	15 FMG/BJN	16 PTN/PWC	17 ASS/CHI
18	19 PLD.FCB	20 BJN/CSM	21 PWC/GSW	22 CHI/TOR	23 AAL/LAL	24 FMG/ZLK
25	26 PTN/SDK	27 ASS/NZB	28 PLD/GST	29 BJN/MCF	30 PWC/LFL	31 CHI/SMB
Sep.1	2 FCB/AAL	3 CSM/FMG	4 GSW/PTN	5 TOR/ASS	6 LAL/PLD	7 ZLK/BJN
8	9 SDK/PWC	10 NZB/CHI	11 GST/FCB	12 MCF/CSM	13 LFL/GSW	14 SMB/TOR
15	16 FCB/LAL	17 CSM/ZLK	18 GSW/SDK	19 TOR/NZB		

Stage 1: Group Stage

<i>G1</i>	<i>HOME</i> <i>AWAY</i>	Los Angeles Lakers	FC Barcelona	Petro de Luanda	Al Ahly	Guangdong Southern Tigers	<i>G3</i>	<i>HOME</i> <i>AWAY</i>	Golden State Warrior	Panathinaikos Athens	Sydney kings	Perth Wildcats	Liaoning Flying Leopards
	LAL		25.11.13 2:00 PM	25.11.23 2:00 PM	25.12.3 1:00 PM	25.12.12 7:00 AM		GSW		25.12.16 12:00 PM	25.10.15 5:00 AM	25.10.23 7:00 AM	25.12.7 7:00 AM
	FCB	25.10.9 11:00 PM		25.12.2 2:00 PM	25.11.22 0:30 PM	25.12.20 7:00 AM		PTN	25.11.1 11:00 PM		25.12.6 5:00 AM	25.11.28 7:00 AM	25.11.18 7:00 AM
	PLD	25.10.19 11:00 PM	26.1.6 2:00 PM		25.10.1 1:00 PM	25.10.10 6:00 AM		SDK	25.11.27 11:00 PM	26.1.1 12:00 PM		25.12.17 6:00 AM	25.10.24 6:30 AM
	AAL	25.12.21 10:30 PM	25.12.11 2:00 PM	25.10.29 1:30 PM		26.1.7 7:00 AM		PWC	25.11.19 10:30 PM	25.10.5 11:00 PM	26.1.11 5:00 AM		25.11.2 7:00 AM
	GST	25.9.30 11:00 PM	25.10.28 2:00 PM	25.11.14 2:00 PM	25.10.18 1:00 PM			LFL	26.1.2 11:00 PM	26.1.10 12:00 PM	25.10.4 5:00 AM	25.10.14 7:00 AM	
<i>G2</i>	<i>HOME</i> <i>AWAY</i>	Chigago Bulls	Toronto Raptors	San Miguel Beermen	AS Sale	Breakers	<i>G4</i>	<i>HOME</i> <i>AWAY</i>	Real Madrid	CSKA Moscow	Zamalek	Flamengo	Boca Juniors
	CHI		25.11.25 8:00 PM	25.10.12 7:00 AM	25.10.3 3:00 PM	25.12.4 2:30 AM		MCF		25.11.30 12:00 PM	25.10.6 1:00 PM	25.10.25 6:00 PM	25.11.12 5:00 PM
	TOR	26.1.9 9:00 PM		25.10.31 7:00 AM	25.10.21 3:00 PM	25.11.15 3:30 AM		CSM	26.1.4 2:00 PM		26.1.14 1:00 PM	25.11.11 6:00 PM	25.11.20 6:00 PM
	SMB	25.12.30 9:00 PM	25.12.13 8:00 PM		25.12.5 3:00 PM	25.10.2 3:00 AM		ZLK	25.12.9 2:00 PM	25.10.17 12:00 PM		25.12.19 5:30 PM	26.1.3 6:00 PM
	ASS	25.12.14 8:30 PM	25.12.31 7:30 PM	25.11.16 7:00 AM		25.11.26 3:00 AM		FMG	25.11.21 2:00 PM	25.12.10 11:30 PM	25.11.29 1:00 PM		26.1.13 6:00 PM
	NZB	25.10.22 9:00 PM	25.10.11 8:00 PM	26.1.8 7:00 AM	25.10.31 2:30 PM			BJN	25.10.16 1:30 PM	25.12.18 12:00 PM	25.10.26 1:00 PM	25.10.8 6:00 PM	

Jan.19	20	21	22	23	24	25	Break
26 Upper Bracket G1	27	28 Lower Bracket G1	29	30 Upper Bracket G2	31	Feb.1 Lower Bracket G2	Stage 2 2
2	3	4 Last Chance G1	5	6	7 Last Chance G2	8	
9	10 Final Twelve S1 G1	11 Final Twelve S2 G1	12	13 Final Twelve S3 G1	14	15 Final Twelve S4 G1	Stage 3 3
16 Final Twelve S1 G2	17 Final Twelve S2 G2	18	19 Final Twelve S3 G2	20	21 Final Twelve S4 G2	22	
23	24 Final Twelve S1 G3*	25 Final Twelve S2 G3*	26	27 Final Twelve S3 G3*	28	Mar.1 Final Twelve S4 G3*	
2	3	4	5 Final Eight S1 G1	6	7 Final Eight S2 G1	8 Final Eight S3 G1	Stage 4 4
9	10 Final Eight S4 G1	11 Final Eight S1 G2	12	13 Final Eight S2 G2	14 Final Eight S3 G2	15	
16 Final Eight S4 G2	17	18 Final Eight S1 G3*	19	20 Final Eight S2 G3*	21 Final Eight S3 G3*	22	
23 Final Eight S4 G3*	24	25	26	27 Quarterfinal S1 G1	28 Quarterfinal S2 G1	29	
30 Quarterfinal S1 G2	31 Quarterfinal S2 G2	Apr.1	2 Quarterfinal S1 G3	3 Quarterfinal S2 G3	4	5	
6	7 Quarterfinal S1 G4*	8 Quarterfinal S2 G4*	9	10 Quarterfinal S1 G5*	11 Quarterfinal S2 G5*	12	
13	14	15	16 Final G1	17	18 Final G2	19	
20 Final G3	21	26 Final G4	23	246 Final G5*	25	26 Final G6*	
28	27 Final G7*	Notes: S#: Series# G#: Game# (S# G#)*: The Game Played if needed					

References

- [1] [Online]. Available: <https://www.forbes.com/>.
- [2] R. W. Saaty, “The analytic hierarchy process—what it is and how it is used,” *Mathematical modelling*, vol. 9, no. 3-5, pp. 161–176, 1987.
- [3] K. Devriesere, L. Csató, and D. Goossens, “Tournament design: A review from an operational research perspective,” *European Journal of Operational Research*, 2024.
- [4] Dec. 2024. [Online]. Available: https://en.wikipedia.org/wiki/Round-robin_tournament.
- [5] *Swiss-system tournament - Wikipedia — en.wikipedia.org*, https://en.wikipedia.org/wiki/Swiss-system_tournament, [Accessed 11-03-2025].
- [6] Mar. 2025. [Online]. Available: https://en.wikipedia.org/wiki/FIFA_World_Cup.
- [7] *Major league soccer*, Mar. 2025. [Online]. Available: https://en.wikipedia.org/wiki/Major_League_Soccer.
- [8] T.-K. Huang, R. C. Weng, C.-J. Lin, and G. Ridgeway, “Generalized bradley-terry models and multi-class probability estimates,” *Journal of Machine Learning Research*, vol. 7, no. 1, 2006.
- [9] K. G. Edwards, “The biggest game in town: The nfl, the epl, and a new model for value creation in professional sports,” Ph.D. dissertation, University of Oregon, 2013.
- [10] J. H. Dinitz and V. Burlington, “Designing schedules for leagues and tournaments,” *Graph Theory Notes of New York*, vol. 48, pp. 7–21, 2005.
- [11] P. Barčák, “Application of the hooper index in monitoring training load and player well-being: A theoretical review on performance optimization in soccer,” *Timisoara Physical Education & Rehabilitation Journal*, vol. 17, no. 33, 2024.
- [12] *League of Legends MSI 2022: RNG’s games to be replayed after opponents’ ping, latency issues — sg.news.yahoo.com*, <https://sg.news.yahoo.com/league-of-legends-msi-2022-rng-games-replay-ping-issues-103306979.html?guccounter=1>, [Accessed 11-03-2025].
- [13] Y. Chikish, M. Carreras, and J. García, “Esports: A new era for the sports industry and a new impulse for the research in sports (and) economics,” *Sports (and) economics*, pp. 477–508, 2019.

Appendices

Appendix A Disclosure of Generative AI Usage

Use of LLM: We used the LLMs to generate part of our essay. Deepseek R1 and Chatgpt 4.0 were used to write the essay and search for data. Chatgpt 4.0 was also used to search academic papers.

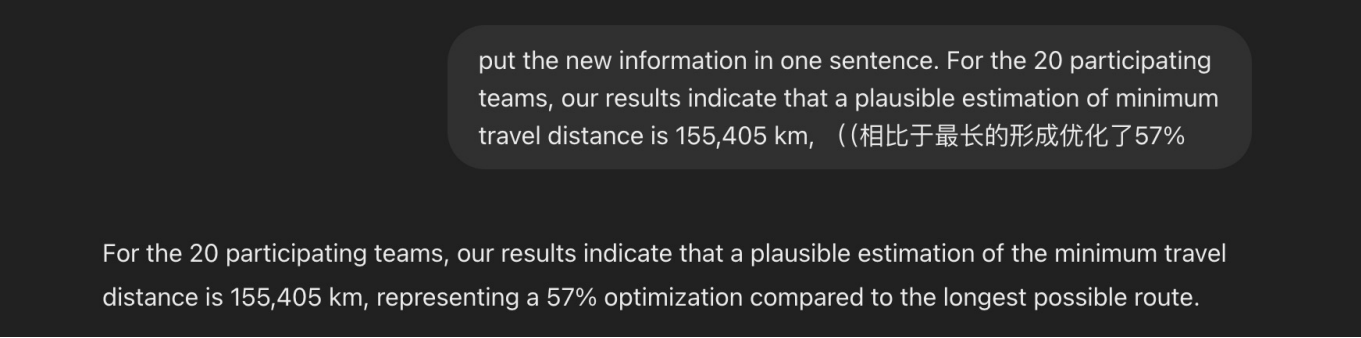


Figure 5: Chapgpt 4o



Figure 6: Deepseek